

Group 2

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Multiplayer Blackjack **Implementation**



Introduction



Translating Design

From ideas to canvas, bringing the design to life

Implementation & Testing

Did our vision get realized? Did our tests and work reflect that?

How did we deliver?

Did we provide on our promise?

Networking



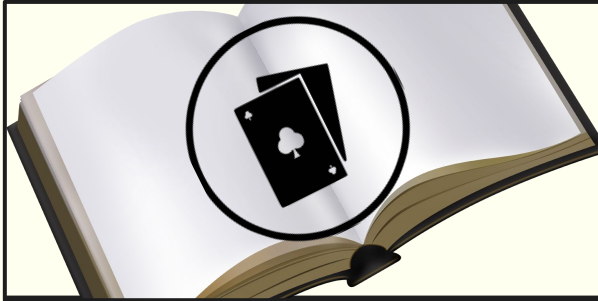
- Hardest piece to get work working, and bottleneck for other parts of the project.
- Successfully implemented
 - Singleton
 - Server is a single source of truth.
 - Prevents other versions of the class from running within this current instance of the application
 - Multithreaded Socket
 - Allows for Multiple clients to connect simultaneously and for the server
 - Object over Socket
 - Server side messaging setup to translate any client message into an action a reply with a corresponding message
 - Client side messaging servers messages into client state.

Networking



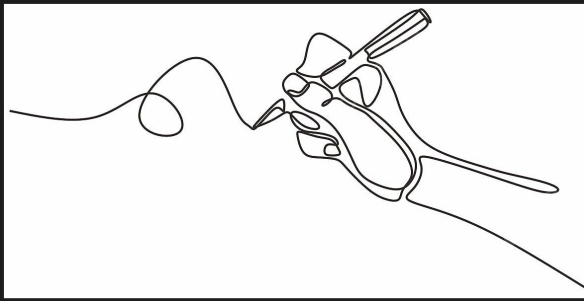
- Network Implementation issues:
 - Getting the Client side listener and sent thread to work together
- Unexpected networking issues
 - Broadcasting message to multiple users simultaneously
 - Needs to be done server wide to announce tables to each client
 - Need to create subgroups for broadcast messages on a per table basis to announce other player and dealer actions.

Game Logic



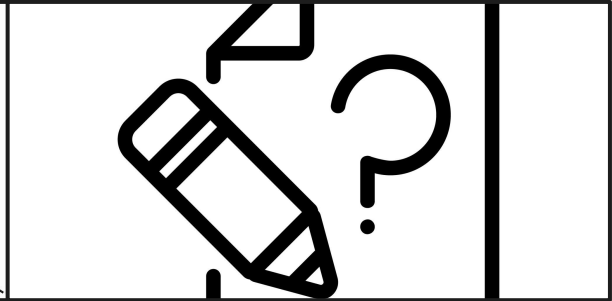
Blackjack Concepts

- Translation of known ideas to technical processes
- Functionally making Blackjack work
- Challenges in covering cases and conditions



Design to Code

- Initial expectations needed revisions to become transferable to code
- New knowledge and obstacles led to various design changes
- Bridging the gap between modules



JUnit Testing

- Quickly became important to the goal of reliable game functionality
- Automation/varied test cases were unexpectedly major bonuses

Game Logic Cont.

Design is Key

- Major lesson from making a design to mapping out the program

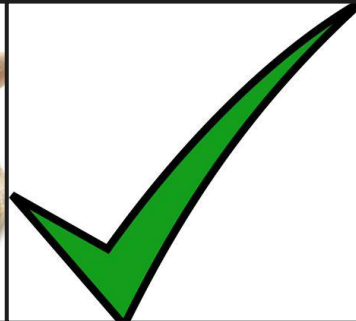


Game Dev Prior Experience versus Building From Scratch

- Valuable learning moment



JUnit Testing Successes



Java CI

✓ Merge pull request #26 from eb-heavybarrel/n1ckvu-patch-2 #111

Summary

All jobs

Build and Run Tests

Run details

Usage

Workflow file

Build and Run Tests

succeeded 1 hour ago in 13s

- > ✓ Set up job
- > ✓ Checkout Code
- > ✓ Setup JDK
- > ✓ Download JUnit JAR
- > ✓ Compile Java code
- > ✓ Copy properties file
- > ✓ Run JUnit tests
- > ✓ Upload Test Report Artifact
- > ✓ Post Setup JDK
- > ✓ Post Checkout Code
- > ✓ Complete job

GUI Mock-ups



Login

Username

Password

IP

>>

>Create Account

****Login Success/Failure****

Tables			Add Funds/ Cash Out	Rules	Sign Out
●	Table 1	Turn: 45s Bet: 100	*view funds + fund actions *rule display *sign out confirmation		
●	Table 2	Turn: 45s Bet: 25			
●	Table 3	Turn: 15s Bet: 10			
●	Table 4	Turn: 45s Bet: 650			
●	Table 5	Turn: 25s Bet: 850			
●	Table 6	Turn: 30s Bet: 200			
●	Table 7	Turn: 20s Bet: 50			
■					
■					

0:45

Player 1 Score: XXX			Hit Stand Surrender Double-Down Bet All in
Player 2 Score: XXX			
Player 3 Score: XXX			
Player 4 Score: XXX			
Player 5 Score: XXX			
Player 6 Score: XXX			

GUI - Login



Blackjack Login

Login

Username	
Password	
IP	

>>

>Create Account

****Login Success/Failure****

Blackjack Login

Create Account

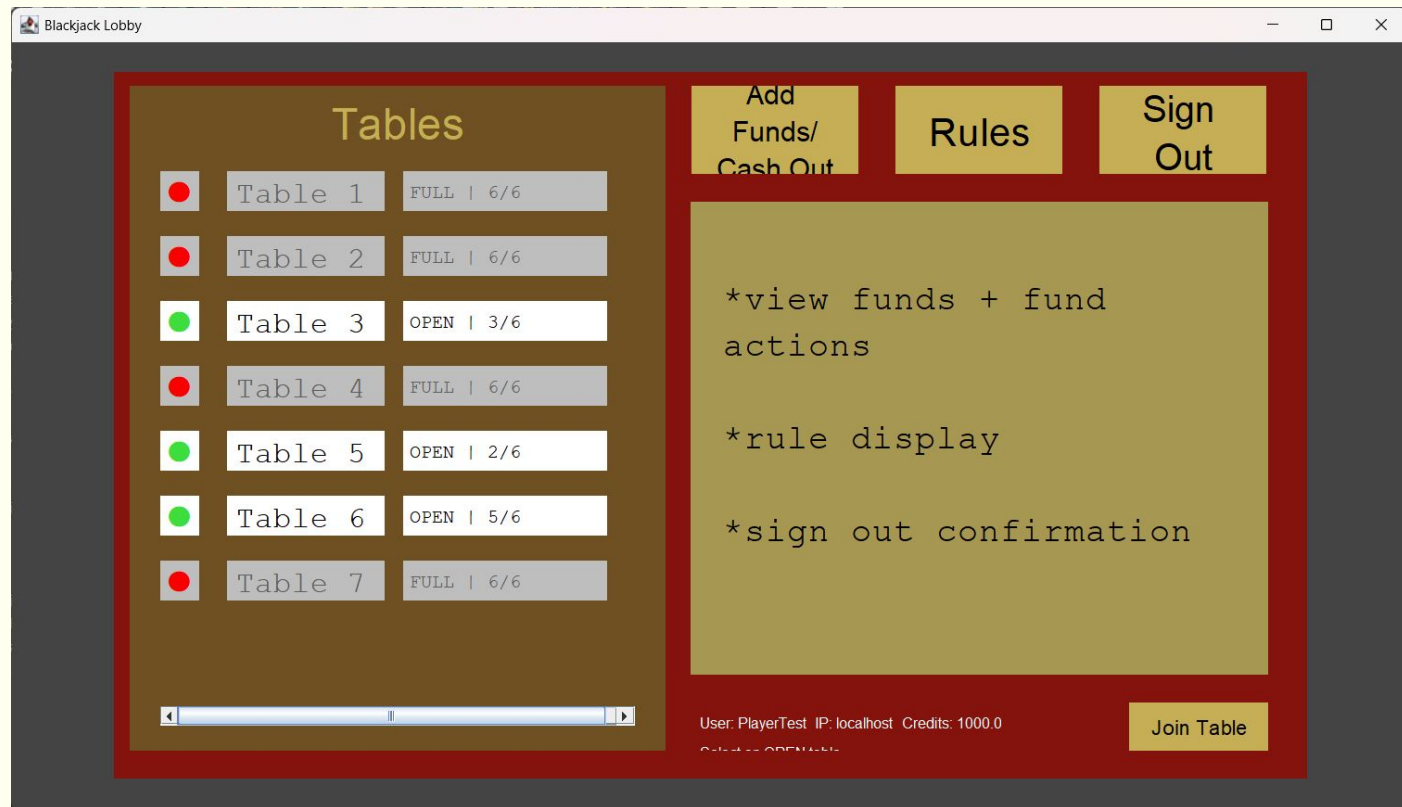
Username	
Password	
Confirm Password	
IP	

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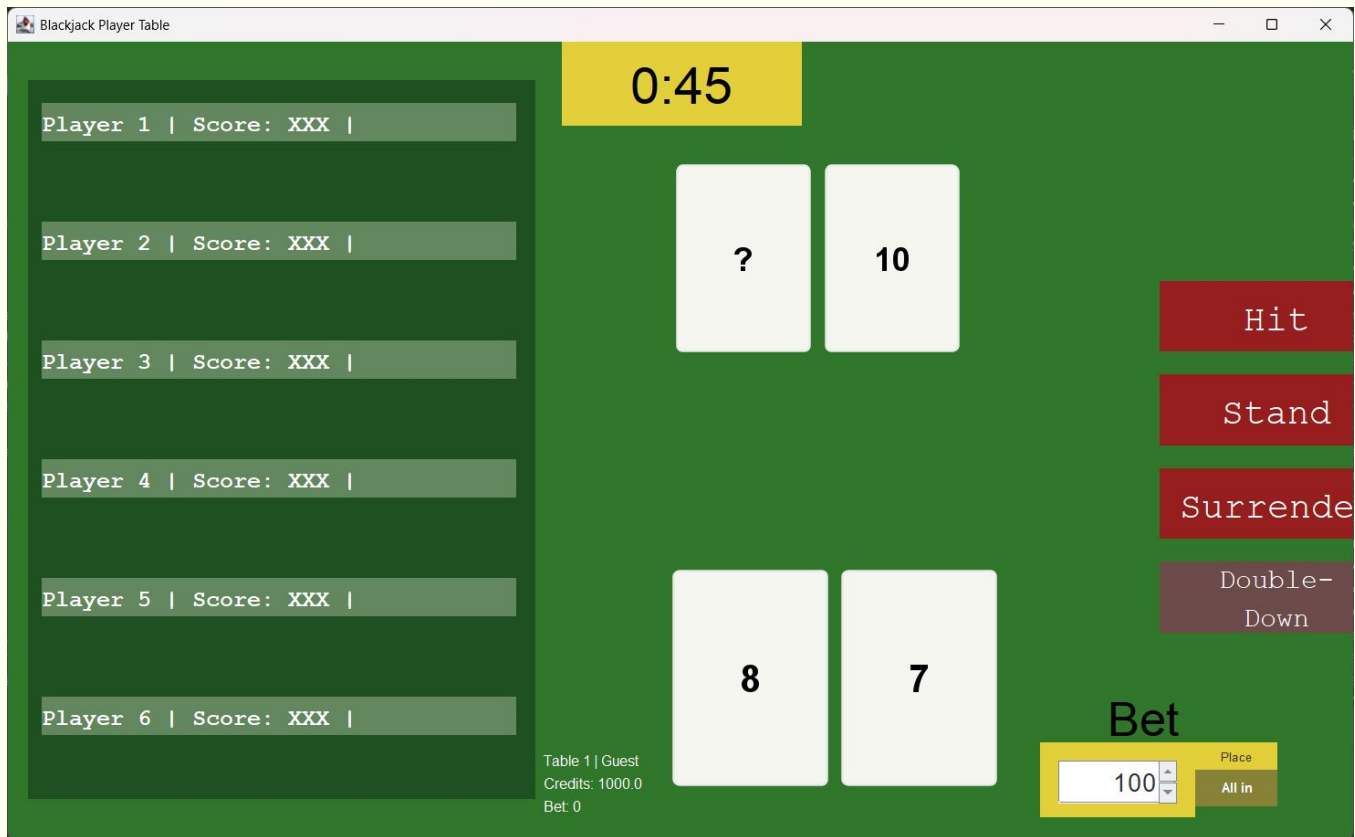
<Back to Login

Please enter a unique username and strong password

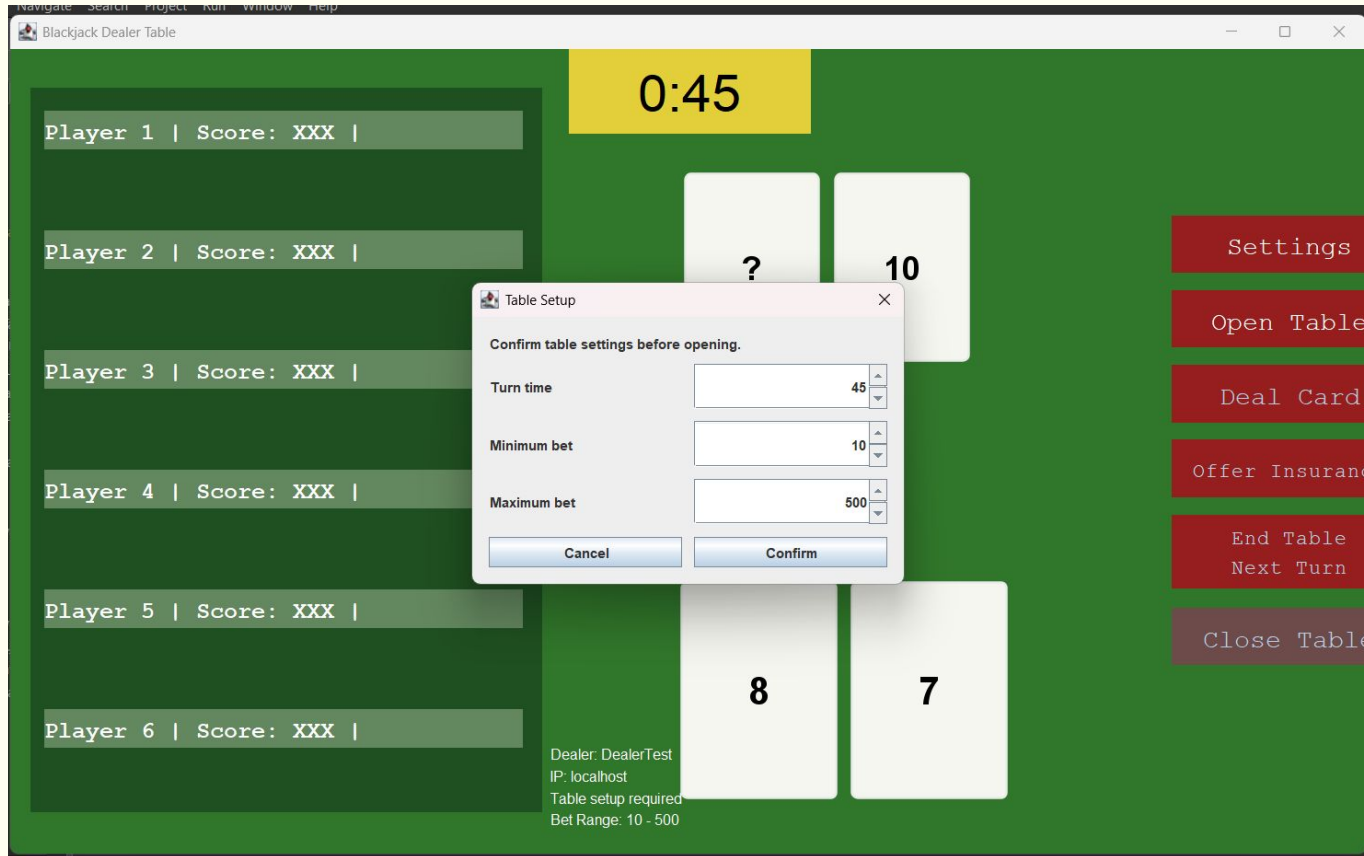
GUI - Lobby



GUI - Player



GUI - Table (Dealer)



Closing & Retrospective

- From CS401 - Successfully implementing design patterns, tackling program design
- Outside CS401 - synchronizing, broadcast messages, research considerations for modeling the game
- UI/UX Perspective + Scope
- Implementation Issues
- Design Changes and Shifting Priorities



```
Run JUnit tests
58 |
59 | └─ JUnit Platform Suite ✓
60 |   └─ All Tests ✓
61 |     └─ JUnit Jupiter ✓
62 |       └─ UserTest ✓
63 |         └─ testIncrementCredits() ✓
64 |         └─ testSaveUser() ✓
65 |         └─ testLogin() ✓
66 |         └─ testGetRole() ✓
67 |         └─ testChangePassword() ✓
68 |         └─ testDecrementCredits() ✓
69 |         └─ getLoginStatus() ✓
70 |         └─ testSetUserName() ✓
71 |         └─ testGetUserName() ✓
72 |         └─ testGetCredits() ✓
73 |         └─ testLogout() ✓
74 |         └─ testSetRole() ✓
75 |       └─ BlackjackServerTest ✓
76 |         └─ test() ✓
77 |       └─ MessageStatusTest ✓
78 |         └─ test() ✓
79 |       └─ MessageTest ✓
80 |         └─ testGetmClass() ✓
81 |         └─ testGetData() ✓
82 |         └─ testGetType() ✓
83 |         └─ testGetmStatus() ✓
84 |       └─ TableTest ✓
85 |         └─ testFirstJoinEmptySeat() ✓
86 |         └─ testTableIsFull() ✓
87 |         └─ testTableInsertion() ✓
88 |         └─ testPlayerRemoval() ✓
89 |       └─ GameTest ✓
90 |         └─ testEndConditionPlayerBlackjack() ✓
91 |         └─ testDealerEqualToPlayer() ✓
92 |         └─ testDealerHandGreaterThanPlayer() ✓
93 |         └─ testBust() ✓
94 |         └─ testEndConditionDealerBlackjack() ✓
95 |         └─ testDealer21ToPlayer21() ✓
96 |         └─ testEndConditionPush() ✓
97 |         └─ testPlayerHandGreaterThanDealer() ✓
98 |       └─ HandTest ✓
```